

Omid Arefian

Robotics and State Estimation Engineer

Milan, Italy | +39 366 141 7872

Omid.arefian@mail.polimi.it

linkedin.com/in/omid-arefian | github.com/OmidArefian01

Portfolio: omidarefian01.github.io



Professional Summary

M.Sc. Mechanical Engineering candidate in Mechatronics and Robotics with experience developing real-time state-estimation, sensor-fusion, computer-vision, and autonomous-navigation systems. Current thesis work at SunCubes focuses on UAV localization using a multiplicative extended Kalman filter (MEKF), monocular pose estimation, adaptive measurement covariance, PX4 integration, and federated Kalman filtering. Proficient in Python, C++, MATLAB/Simulink, ROS 2, PX4, MAVLink, OpenCV, PyTorch, and statistical consistency analysis using innovations, NIS, NEES, and RMSE.

Professional Experience

Estimation Engineer, Thesis Internship

Mar 2026–Present

SunCubes

Milan, Italy

- Characterized IMU noise using overlapping Allan deviation, Hadamard deviation, Welch power spectral density, time-domain residual analysis, and datasheet comparison. Distinguished short-term white noise from bias instability, stochastic random walk, temperature effects, and deterministic drift.
- Translated the experimentally supported sensor-noise parameters into the MEKF process-noise covariance Q , while separating measured parameters from provisional values that require longer temperature-controlled recordings.
- Developed a Multiplicative Extended Kalman Filter (MEKF) for UAV position, velocity, quaternion attitude, and online sensor-bias estimation using IMU propagation and corrections from barometer, rangefinder, optical flow, and magnetic heading.
- Identified sensor measurement covariance R using stationary recordings and flight-data replay. Evaluated estimator consistency through innovations, Normalized Innovation Squared (NIS), measurement-rejection rates, covariance health, and trajectory behaviour.
- Developed KFPnP, a monocular ArUco pose-estimation pipeline combining OpenCV marker localization, a PyTorch CNN for robust marker-ID and rotation recovery, camera calibration, and covariance-weighted maximum-likelihood PnP.
- Designed an adaptive image-noise model that estimates an independent 2×2 covariance matrix for every detected marker corner from local image gradients, illumination, and camera-noise characteristics.
- Propagated the adaptive corner uncertainty through the PnP solution to obtain a complete 6×6 pose covariance, enabling the Kalman filter to adjust its measurement covariance R_k according to the quality of each camera frame.
- Validated KFPnP using three simultaneously tracked markers under continuous illumination changes. The estimated uncertainty increased by approximately $2.5\times$ from the brightest to the darkest conditions, demonstrating that the covariance model responds to image degradation.
- Used the KFPnP pose and covariance as adaptive measurements for temporal Kalman filtering, allowing reliable frames to produce stronger corrections while uncertain frames are down-weighted and short visual dropouts are handled through prediction.
- Integrated the inertial MEKF and vision-based Kalman filter through a federated Kalman-filter architecture for continuous UAV tracking. Deployed the estimator in PX4 using C++ and transmitted synchronized outputs through a custom MAVLink dialect to a Raspberry Pi 5 companion computer.

Research Assistant, Part-time

Jun 2023–Feb 2024

Shahid Beheshti University

Tehran, Iran

- Supported experimental measurement campaigns and characterization of photovoltaic-panel technologies in the Energy Engineering Laboratory.
- Co-authored six conference posters; one related paper was peer reviewed and accepted for international presentation.

Mechanical Engineering Intern

Jun 2022–Sep 2022

Khorasan Steel Complex

Iran

- Supported inspection, maintenance, and troubleshooting activities on the rebar rolling-mill repair line and gained practical experience with industrial machinery and maintenance workflows.

Education

M.Sc. Mechanical Engineering — Mechatronics and Robotics

Politecnico di Milano

2024–Present

Milan, Italy

Thesis: Federated Kalman Filtering for UAV Tracking Using Vision and Multi-Sensor Fusion. GPA: 25.21/30.

B.Sc. Mechanical Engineering

Shahid Beheshti University

2019–2024

Tehran, Iran

GPA: 17.77/20. Full merit scholarship.

Selected Projects

KFPnP: Robust and Adaptive ArUco Pose Estimation

2026

Python, OpenCV, PyTorch, CNN, MLPnP, camera calibration, Kalman filtering

- Built a real-time pipeline that recovers ArUco ID and orientation under blur, glare, noise, low light, and viewpoint changes using a 201-class CNN operating on normalized 40×40 crops.
- Estimated per-corner covariance from sensor noise and local image gradients, then used covariance-weighted MLPnP to produce a 6-DoF pose and pose covariance.
- Added adaptive- R temporal tracking with innovation monitoring, NIS diagnostics, Joseph-form covariance correction, and prediction through short visual dropouts.

SLAM-Based Navigation and Path Planning on TurtleBot3

2025

ROS 2, Gazebo, RViz2, SLAM, A*, AMCL, pure pursuit

- Implemented an end-to-end navigation stack covering occupancy-grid mapping, global planning, particle-filter localization, waypoint tracking, and laser-scan obstacle avoidance.
- Benchmarked five search variants; A* with a Manhattan heuristic explored 44% fewer nodes than Dijkstra while returning an equal-length path and completed the representative search in 0.0103 s.

Nonlinear Optimal Control and Trajectory Generation

2025

Python, MATLAB, Pontryagin minimum principle, direct shooting, LQR

- Formulated obstacle-avoiding trajectory optimization for a nonlinear 2D vehicle and compared indirect PMP shooting with direct single shooting.
- Derived state and costate dynamics, implemented forward–backward optimization, and designed time-varying LQR tracking around the optimized trajectory.

Custom PPO Environment for Multi-Box Coordination

2025

Python, reinforcement learning, PPO, GAE, reward shaping

- Built a custom environment for collision-aware multi-box coordination and evaluated entropy regularization, reward shaping, and three neural-network architectures.
- The [1024, 512, 256] network achieved the best reported final reward (-1.10×10^6), compared with -1.15×10^6 and -1.33×10^6 for the other tested architectures; an entropy coefficient of 0.05 produced the strongest final moving-average reward.

Experimental Dynamics and System Identification

2025

MATLAB, frequency-response functions, modal analysis, finite-element correlation

- Performed spectral FRF testing on a steel plate using H1/H2 estimation, finite-element correlation, and MDOF-to-SDOF reduction; conducted free–free modal analysis of a light-rail wheel.

Technical Skills

Programming: Python, C++, MATLAB, Simulink, Git, Linux.

State Estimation: Kalman filter, EKF, MEKF, UKF, error-state estimation, federated filtering, GNSS/INS integration, covariance propagation, innovation gating, NIS/NEES, RMSE analysis.

Computer Vision and Machine Learning: OpenCV, PyTorch, CNN training, ArUco, PnP/MLPnP, camera calibration, image processing, synthetic data, uncertainty modelling.

Robotics and Embedded Systems: ROS 2, PX4, MAVLink/PyMAVLink, Gazebo, RViz2, Raspberry Pi, SLAM, AMCL, path planning, autonomous navigation.

Control and Modelling: MPC, LQR, PID, nonlinear control, optimal control, system identification.

Engineering Tools: ANSYS, SolidWorks.

Languages: Persian (native), English (C1), Italian (A2).

Awards

2025: Second place, SkyEU Hackathon, Milan.

2019–2024: Full bachelor's scholarship, Shahid Beheshti University, academic merit.

2019: Top 1% nationally, Iranian National University Entrance Examination.